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DSC01
ALL

# Jharkhand University of Technology, Ranchi

## Diploma 1st Semester Examination, 2023 (NEP)

**Subject : Engineering Physics** **Subject Code : DSC01**

**Time Allowed : 3 Hours** **Full Marks : 70**

*Answer in your own words.*
*Answer* any five *questions.* Question No. 1 *is compulsory.*
*Marks are given in the right margin.*

1. Choose the correct answer in the following: 1×10=10

(i) Which of the following is a vector quantity?

(a) Energy (b) Impulse
(c) Temperature (d) Density

(ii) In the measured length of 0.05060m, the number of significant figures is

(a) 5 (b) 4
(c) 2 (d) 3

(iii) A rocket works on the principle of conservation of

(a) mass (b) energy
(c) linear momentum (d) angular momentum

(iv) Newton's first law of motion gives the concept of

(a) work (b) power
(c) momentum (d) inertia

(v) The weight of a body at the centre of the earth is

(a) zero (b) infinite
(c) same as on the surface of earth (d) None of these

(vi) Heat travels through vacuum by

(a) conduction (b) convection
(c) radiation (d) Both (a) and (b)

(vii) Spring is made of steel instead of copper because

(a) steel is cheap. (b) steel is in abundance.
(c) steel is more elastic than copper. (d) None of these

3.71
3.69
/02

3.71
3.70
3.67
3) 11.08 (3.693
9
20
18
28
27

(viii) Pressure has the same dimensional formula as that of

(a) power (b) force

(c) stress (d) strain

(ix) In a stationary wave, the distance between the nearest node and antinode is

(a) $\lambda$ (b) $\frac{\lambda}{2}$

(c) $\frac{\lambda}{4}$ (d) $2\lambda$

(x) The ratio of co-efficient of superficial expansion and co-efficient of linear expansion is

(a) 1 : 1 (b) 2 : 1

(c) 3 : 1 (d) None of these

**2.** (a) What do you mean by dimensions of a physical quantity? Explain with examples. Also write the dimensional formula of momentum, impulse and universal gravitational constant (G).

(b) Diameter of a pipe was measured by Vernier Callipers. The measurements were 3.71 cm. 3.70 cm. and 3.67 cm. Calculate mean absolute error and percentage error. 10+5

**3.** (a) State Newton's laws of motion. Also define force, momentum and impulse. Write their units too.

(b) Explain why a passenger sitting on a running bus tends to fall forward, when the bus suddenly stops. 10+5

**4.** (a) Define the terms — velocity and acceleration. Also derive the following equations for a uniformly accelerated motion.

(i) $v = u + at$

(ii) $s = ut + \frac{1}{2}at^2$

(iii) $v^2 = u^2 + 2as$

where the symbols have their usual meanings.

(b) A car initially at rest starts moving with a constant acceleration of 0.5 $ms^{-2}$ and travels a distance of 25 m. Find :

(i) its final velocity

(ii) the time taken 10+5

**5.** (a) Define acceleration due to gravity($g$) and write its S.I. unit. How is it related to universal gravitational constant($G$)? Explain the variation of '$g$' with altitude and depth.

(b) Define — stress, strain and elastic limit. Also write their S.I. units. 10+5

3) 0.05 (0.018

3.5/2.0 1.5/2

16×10 / 369

$Nm^2/kg$

$ML^3T^{-2}$



6. (a) State the laws of thermal conductivity of a material of a body. Also define the co-efficient of thermal conductivity and write its S.I. unit.

   (b) Describe the various modes of transmission of heat giving one example of each. 10+5

7. (a) What do you mean by longitudinal wave and transverse wave? Also define — stationary wave, node and antinode.

   (b) The audible range of human ear is 20 Hz – 20 kHz. Convert this into the corresponding wavelength range. Take the speed of sound in air at ordinary temperature = 340 $ms^{-1}$. 10+5

$u^2 + 2a(ut + \frac{1}{2}at^2)$

$u^2 + 2uat + a^2t^2$